Selective Classification and Probability Calibration for Trustworthy Stress Detection from Wrist-Worn Physiological Signals

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***Abstract*—Wrist-worn continuous stress monitoring systems offer a non-invasive tool for proactive health management. However, photoplethysmography (BVP) and electrodermal activity (EDA) sensors placed at the wrist are susceptible to movement artifacts. Routine physical activity (walking, desk work, arm articulation) naturally activates the sympathetic nervous system, increasing heart rate and sudomotor sweating in a manner that closely resembles physiological responses to mental stress. Standard machine learning classifiers forced to make predictions on 100% of data windows (Full Coverage) produce high rates of motion-induced false alarms, contributing directly to alarm fatigue. In this paper, we propose a reliable stress detection framework integrating Isotonic probability calibration with selective classification augmented by kinematic context on the benchmark WESAD dataset ($N = 15$). The framework allows the classifier to abstain from making a decision (assigning an *Uncertain* label) when posterior confidence falls below a safety threshold. Systematic benchmarks across eight model architectures (seven tabular classifiers and a multi-layer perceptron) under a 15-fold Leave-One-Subject-Out (LOSO) protocol with inner subject-grouped cross-validation show that motion vulnerability is common across all models ($\text{FAR}_{\text{motion}} \approx 12.2\% - 24.0\%$). The proposed rejection rule suppresses between 55.0% and 90.0% of these motion-induced false alarms across benchmarks. At the optimal confidence threshold $\tau = 0.80$ using XGBoost, the framework achieves a Selective Macro F_1-score of 95.20% (Stress $F_1 = 93.15\%$, 95% CI: [84.80%, 98.54%]), reduces Expected Calibration Error (ECE) to 0.0290, and limits $\text{FAR}_{\text{motion}}$ to 4.07% (95% CI: [0.0%, 11.07%]) while maintaining 69.42% usable coverage (95% CI: [58.34%, 79.84%]). Fold-by-fold breakdowns across the 15 WESAD subjects, Wilcoxon signed-rank tests, tabular-versus-neural comparisons, and TreeSHAP feature attributions demonstrate empirical consistency and examine individual autonomic dissociation on an outlier hypo-responder subject (S14).**

***Index Terms*—Wearable Stress Detection, WESAD Dataset, Selective Classification, Probability Calibration, Empatica E4, Motion-Induced False Alarms, Trustworthy AI, Tabular vs Deep Learning, Leave-One-Subject-Out Evaluation.**

I. Introduction

Prolonged psychological stress is a recognized risk factor for immune dysfunction, metabolic disorders, and cardiovascular complications [1]–[3]. The widespread availability of consumer smartbands and smartwatches allows continuous, non-invasive tracking of physiological parameters in daily life [4]–[6].

However, practical deployment in ambulatory environments faces a persistent obstacle: *motion-induced false alarms* [7]–[9]. Physical movement stimulates the sympathetic nervous system (SNS) to support metabolic demands, causing peripheral vasoconstriction, elevated pulse rate, and eccrine sweat secretion on the palmar and wrist surfaces [10]. When recorded by optical blood volume pulse (BVP) and galvanic skin conductance (EDA) sensors, these physiological adjustments closely resemble the autonomic changes induced by acute psychological stress.

In existing studies on the standard WESAD dataset [5], [11]–[15], models are commonly evaluated under a forced-prediction scheme requiring binary classifications (Stress vs. Non-Stress) over 100% of temporal windows (Full Coverage). During physical activity, both neural networks and standard classifiers often output overconfident erroneous predictions [16]. The resulting high frequency of false warnings during routine physical movement causes alarm fatigue, leading users to disable alerts or stop wearing the monitor.

To address this challenge, our work applies the principles of **Trustworthy Machine Learning**: rather than forcing predictions on all inputs, an ambulatory monitoring system should quantify its predictive uncertainty and withhold decisions on ambiguous or motion-corrupted samples through selective classification [17]–[21].

A. Scientific Contributions

This paper presents four main contributions:

1) **Formalization and Quantification of Motion Vulnerability ($\text{FAR}_{\text{motion}}$):** We define the $\text{FAR}_{\text{motion}}$ metric to isolate model behavior during active motion and benchmark eight distinct algorithms (seven tabular models and a multi-layer perceptron) under 15-fold Leave-One-Subject-Out (LOSO) cross-validation [22], [23]. We show that motion false alarm vulnerability affects all model families ($\text{FAR}_{\text{motion}} \approx 12.2\% - 24.0\%$) when full coverage is enforced.
2) **Dual-level Group-aware Calibration Pipeline:** We design an Isotonic probability calibration protocol nested inside an inner subject-stratified cross-validation (`GroupKFold`). This structure prevents data leakage arising from 91.7% sliding-window overlap, aligning model posterior confidence with observed empirical frequencies (ECE $= 0.0290$).

3) **Selective Decision Framework with Kinematic Context Integration:** We formulate an abstention rule based on confidence threshold τ that leverages tri-axial accelerometry (ACC) features. The framework reduces motion-induced false alarms by 55.0% to 90.0% across all eight evaluated models, enabling calibrated XGBoost to reach a Selective Macro F_1 of 95.20% at 69.42% coverage, validated by multi-seed tests (5 seeds) and Wilcoxon signed-rank tests.
4) **Biomedical XAI Attribution and Tabular-vs-Deep Empirical Analysis:** Using TreeSHAP [24], we quantify the relative contributions of four sensor modalities and identify the conflicting feature patterns that trigger model abstention. We provide a controlled empirical comparison between tree ensembles and neural models on the same standardized feature space, analyzing an outlier participant with muted electrodermal response (Subject S14) to inform future personalized architectures.

II. Related Work & Research Gap Analysis

A. Wrist-Based Wearable Stress Detection

Early ambulatory stress monitoring focused primarily on chest-worn straps or multi-lead electrocardiography (ECG) [25]. With the introduction of multi-sensor wrist devices such as the Empatica E4 [26], recent studies have emphasized optical BVP, EDA, skin temperature (TEMP), and tri-axial accelerometry (ACC) [3], [4].

Schmidt et al. [5] established the WESAD benchmark ($N = 15$), capturing synchronized chest and wrist data. Their findings indicated that wrist-based classification is considerably more challenging than chest-based monitoring due to unrestricted arm movement. Bobade and Vani [12] examined classical classifiers and deep CNN/LSTM architectures, achieving 75%–84% accuracy on wrist data; however, their evaluation used random K-Fold splitting, which causes temporal data leakage between overlapping windows. Siirtola [11] extracted heart rate variability (HRV) metrics from wrist BVP, noting that individual physiological differences present substantial hurdles for cross-subject generalization. Can et al. [3], Sano et al. [6], and Böttcher et al. [15] corroborated that physiological heterogeneity degrades accuracy in real-world settings. Recent works by Behinaein et al. [14] and Dalmeida and Masala [13] investigated deep transformer and vision-based architectures, though computational demands on embedded hardware remain an open constraint.

B. Motion Artifacts and False Alarms

Arm and wrist movements distort optical BVP waveforms and change skin-electrode contact geometry [7]–[9]. In addition to physical displacement artifacts, physical activity stimulates metabolic and autonomic adjustments: heart rate increases and eccrine sweat glands activate for thermoregulation [7], [10].

Prior research has approached motion primarily through signal processing methods, such as Butterworth filtering or adaptive artifact cancellation [9]. While frequency filtering removes high-frequency electrical and optical noise, it cannot alter authentic physiological tachycardia and sweat production caused by exercise. Consequently, when classifiers are required to predict on all windows, false alarm rates remain high during physical activities, leading to alarm fatigue [3].

C. Selective Classification and Learning with Rejection

Originating with Chow's formulation [17], learning with rejection allows a classifier to withhold judgment when the expected cost of an error exceeds the cost of abstention. Geifman and El-Yaniv [19], [20] adapted selective classification to deep neural networks, analyzing the risk-coverage trade-off. Additional theoretical work by Cortes et al. [18], Grandvalet et al. [27], Wiqvist et al. [28], and a survey by Hendrickx et al. [21] examined rejection frameworks in risk-sensitive applications.

Despite these developments, selective classification has not been widely utilized for wearable stress tracking. Our work applies confidence thresholding τ informed by kinematic context to govern selective abstention, specifically mitigating motion-induced false alarms.

D. Probability Calibration and Windowing Leakage

Effective selective classification requires well-calibrated posterior probabilities: the predicted confidence should reflect the empirical probability of correctness [16]. Uncalibrated models often produce overconfident outputs. Standard post-processing calibration techniques include Platt scaling (Sigmoid) [29], Beta calibration [30], and non-parametric Isotonic regression [31], [32].

In physiological time series, sliding windows (e.g., 60-second windows with 5-second shifts) introduce 91.7% overlap between consecutive observations. Applying conventional random cross-validation allows contiguous windows from the same subject to appear in both training and calibration splits, introducing optimistic bias in calibration metrics [22], [23]. We address this via a two-level evaluation framework with subject-grouped inner splits.

E. Rationale for Tabular Machine Learning

Given the prevalence of deep neural networks, our focus on decision-tree ensembles (Random Forest, LightGBM, XGBoost [33]) is based on three considerations:

1) **Sample Size Constraints ($N = 15$ Subjects):** WESAD contains 15 participants. Studies by Shwartz-Ziv and Armon [34], Grinsztajn et al. [35], and Borisov et al. [36] show that on small-to-medium tabular datasets, tree-based models frequently match or exceed deep neural networks. High-capacity neural networks can overfit subject-specific physiological baselines on small cohorts, impairing leave-one-subject-out generalization.
2) **Biomedical Interpretability (XAI):** Standard HRV and EDA metrics (RMSSD, LF/HF, SCR peaks) are grounded in established autonomic physiology [25],

[37], [38]. Combined with TreeSHAP [24], tree ensembles allow feature attribution for individual predictions and abstentions.

3) **Embedded Hardware Feasibility:** Commercial wrist instruments operate under strict micro-controller power budgets. Decision trees paired with Isotonic look-up tables require low memory and $O(\text{depth})$ inference complexity, facilitating on-device execution.

To evaluate these considerations empirically, we include a Multi-Layer Perceptron (MLP) baseline trained on the same 25-feature space under identical 15-fold LOSO conditions. Table I provides a comparison of recent studies with our proposed methodology.

III. Proposed Methodology

A. Preprocessing and Feature Extraction

Continuous recordings from the Empatica E4 wristband are partitioned into rolling windows of $W = 60\,\text{s}$ with step size $\Delta t = 5\,\text{s}$ (91.7% overlap). A 60-second window length satisfies standard recommendations for short-term HRV estimation [25]:

- **BVP Signal (64 Hz):** Filtered via a 2nd-order Butterworth bandpass filter (0.5–8.0 Hz) to reduce baseline wander. Systolic peaks are identified to compute the inter-beat interval (IBI) sequence. Spurious intervals outside 300–1800 ms are removed, and the resulting series is resampled to 4 Hz via cubic spline interpolation [38]. Extracted metrics include time-domain indicators (Mean HR, SDNN, RMSSD, pNN50) and frequency-domain components via Welch's method: low-frequency power LF (0.04–0.15 Hz), high-frequency power HF (0.15–0.40 Hz), and the sympathovagal balance ratio LF/HF.
- **EDA Signal (4 Hz):** Decomposed using cvxEDA [37] into tonic skin conductance levels (SCL) and phasic skin conductance responses (SCR). Features include SCR peak count, maximum peak amplitude, mean and variance of both tonic and phasic components, and phasic energy $\sum \text{Phasic}^2$ [10].
- **TEMP Signal (4 Hz):** Peripheral skin temperature tracks vasoconstriction. Extracted features include mean, standard deviation, extrema, and linear regression slope.
- **ACC Signal (32 Hz):** Tri-axial acceleration values are combined into the Euclidean norm $a(t) = \sqrt{a_x^2(t) + a_y^2(t) + a_z^2(t)}$. Features include mean, variance $\text{Var}(\text{ACC})$ (proxy for body movement intensity), peak value, zero-crossing rate (ZCR), and spectral energy.

The final dataset comprises 5,294 observation windows across all 15 valid participants from WESAD ($S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S13, S14, S15, S16, S17$; noting that $S1$ was an initial protocol test and $S12$ was excluded by the dataset creators due to hardware sensor malfunction [5]). These 15 participants are evaluated across 15 independent test folds. The standardized feature space consists of 25 continuous features.

B. Probability Calibration

Let $x \in \mathbb{R}^d$ represent the feature vector and $y \in \{0, 1\}$ denote the binary ground truth (0: Non-Stress, 1: Stress). The raw model score $f(x)$ is converted to a calibrated probability via monotonic Isotonic regression [31], [32]:

$$\hat{P}(Y = 1|x) = g(f(x)) \tag{1}$$

where $g(\cdot)$ minimizes squared error subject to non-decreasing monotonicity: $\min_g \sum (y_i - g(f(x_i)))^2$.

Calibration error is evaluated using Expected Calibration Error (ECE) [16] computed over $M = 10$ equal-width bins:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right| \tag{2}$$

where $\text{acc}(B_m)$ and $\text{conf}(B_m)$ denote empirical accuracy and average confidence in bin B_m, respectively.

C. Selective Decision Rule

Let $\hat{p}_1 = \hat{P}(Y = 1|x)$. We define the selective classifier (h, g_τ) with an abstention option governed by confidence threshold $\tau \in [0.5, 1.0)$ [17], [19]:

$$\hat{y}(x) = \begin{cases} 1 \text{ (Stress)}, & \text{if } \hat{p}_1 \geq \tau \\ 0 \text{ (Non-Stress)}, & \text{if } (1 - \hat{p}_1) \geq \tau \\ \text{Uncertain (Abstain)}, & \text{otherwise} \end{cases} \tag{3}$$

Rather than applying a separate heuristic cutoff on acceleration values, our architecture feeds the 6 kinematic features directly into the model alongside the 19 physiological features. Gradient boosted trees learn non-linear relationships between movement patterns and autonomic responses. When intense movement causes conflicting sensor inputs (e.g., elevated heart rate paired with large kinetic variance), the model's calibrated posterior probability naturally settles within the uncertain margin $[1-\tau, \tau]$, triggering abstention without manual thresholding.

Operational coverage ($\mathcal{C}$) indicates the proportion of windows where the classifier outputs a definite prediction:

$$\mathcal{C} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\hat{y}(x_i) \neq \text{Uncertain}) \times 100\% \tag{4}$$

IV. Experimental Setup

A. Dual-Level LOSO Cross-Validation

To assess generalization to unseen individuals, all evaluations follow a 15-fold Leave-One-Subject-Out (LOSO) structure [22], [23]:

- **Level 1 (Outer LOSO Split):** In fold k, all data from the test subject are withheld for independent evaluation. The remaining 14 subjects form the training pool. Feature scaling (`StandardScaler`) is fit exclusively on the 14 training subjects.
- **Level 2 (Group-Aware Inner Calibration):** To prevent leakage across overlapping temporal windows, `CalibratedClassifierCV` uses an inner `GroupKFold(n_splits=3)` grouped strictly by

TABLE I
COMPARISON OF RELATED WEARABLE STRESS DETECTION STUDIES AND THE PROPOSED FRAMEWORK.

Study	Device / Placement	Model Family	Validation Scheme	Probability Calibration	Kinematic Context (ACC)	Rejection Option
Schmidt et al. (2018) [5]	E4 (wrist) + RespiBAN (chest)	LDA, Random Forest, AdaBoost	LOSO (15 folds)	No	Not included	None
Bobade & Vani (2020) [12]	E4 (wrist)	SVM, KNN, CNN, LSTM	Randomized K-Fold	No	Not included	None
Siirtola (2019) [11]	Smartwatch (wrist - BVP)	Random Forest, QDA	LOSO	No	Not included	None
Gjoreski et al. (2016) [4]	Wrist-worn device	Random Forest, Ensemble	In-lab & Real-life	No	Crude activity filter	None
Can et al. (2019) [3]	Multi-sensor wearable	SVM, Gradient Boosting	Multi-protocol	No	Survey analysis	None
Behinaein et al. (2022) [14]	Multi-sensor wearable	Transformer, Deep Learning	LOSO	No	Not included	None
Proposed Study	**Empatica E4 (wrist)**	**8 Models (7 Tabular + MLP)**	**Dual-level LOSO 15-folds**	**Yes (Inner GroupKFold Isotonic)**	**Integrated 6 ACC Features**	**Yes (Selective Rule $\tau = 0.80$)**

`subject_id` across the training participants. This ensures that no individual's windows span both training and calibration folds.

B. Eight-Model Benchmark Suite (7 Tabular ML + 1 Deep MLP)

To examine performance consistency across algorithms, we evaluate eight models: 1) *Logistic Regression (LR)*: Standard linear baseline with L2 penalty; 2) *Decision Tree (DT)*: CART tree (max_depth=5); 3) *k-Nearest Neighbors (kNN)*: Distance-based ($k = 5$); 4) *AdaBoost*: Adaptive boosting ($n_{\text{est}} = 50$); 5) *Random Forest (RF)*: Bagged ensemble ($n_{\text{est}} = 100$, max_depth=5); 6) *LightGBM*: Gradient boosted trees ($n_{\text{est}} = 100$, learning_rate=0.08); 7) *XGBoost*: Extreme gradient boosting ($n_{\text{est}} = 100$, max_depth=5, learning_rate=0.08, subsample=0.8) [33]; 8) *MLP (Deep Learning)*: Multi-layer perceptron (Linear(25, 64) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64, 32) $\rightarrow$ ReLU $\rightarrow$ Linear(32, 2)), optimized via Adam ($lr = 10^{-3}$, weight decay 10^{-4}, 100 epochs with Early Stopping). The MLP provides a standardized parametric neural baseline evaluated on the identical 25-feature tabular input.

C. Statistical Evaluation and Bootstrap Confidence Intervals

- **Wilcoxon Signed-Rank Test:** Two-sided non-parametric test over paired LOSO test folds, reporting test statistic W, exact p-value, and rank-biserial effect size (r_{rb}).
- **95% Bootstrap Confidence Intervals:** Non-parametric cluster-adjusted bootstrapping ($B = 1000$ iterations) over out-of-fold predictions to compute intervals $[\text{CI}_{95}^{\text{low}}, \text{CI}_{95}^{\text{high}}]$.

D. Motion-Induced False Alarm Metric (FAR_{motion})

We utilize $\text{FAR}_{\text{motion}}$ to quantify error rates specifically during physical movement. The movement threshold θ_{motion} is set at the 80th percentile of acceleration variance $\text{Var}(\text{ACC})$ across the corpus ($\theta_{\text{motion}} = 15.385$). The metric measures the proportion of non-stress windows with elevated motion that are incorrectly classified as stress:

$$\text{FAR}_{\text{motion}} = \frac{\sum_i \mathbb{I}(\hat{y}_i = 1,\, y_i = 0,\, \text{Var}_i \geq \theta_{\text{mot}})}{\sum_i \mathbb{I}(y_i = 0,\, \text{Var}_i \geq \theta_{\text{mot}})} \times 100\% \quad (5)$$

where the denominator corresponds to all 295 high-motion non-stress windows in WESAD.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Eight-Model Benchmark Comparison (Answering RQ1)

Table II and Fig. 1 report consolidated Out-of-Fold (OOF) evaluation metrics across all eight model families over 5,294 WESAD windows. All metrics in Table II are computed over the pooled out-of-fold predictions (Aggregate OOF).

Results across the benchmark indicate two primary findings:

1) **Broad Motion Vulnerability:** Under mandatory full coverage (100%), all models exhibit elevated false alarm rates during physical activity: $\text{FAR}_{\text{motion}}$ ranges from 12.20% (XGBoost) to 24.00% (MLP). Although the MLP achieves competitive overall discrimination ($F_1 = 88.95\%$), it also records the highest motion false alarm rate (24.00%). This observation confirms that motion vulnerability is primarily a physiological confound rather than an artifact of a specific algorithm family.
2) **Impact of Selective Abstention:** Applying selective classification at $\tau = 0.80$ reduces motion false alarms across **all eight architectures** (relative reductions between 55.0% and 90.0%), while raising Macro F_1 by 3.6% to 8.0%. XGBoost achieves a balanced operating profile: limiting $\text{FAR}_{\text{motion}}$ to 4.07% (a 66.6% reduction relative to raw baseline) and achieving 95.20% Selective Macro F_1 at 69.42% coverage. Although LightGBM achieves the lowest absolute $\text{FAR}_{\text{motion}}$ (1.36%, a 90.0% reduction), it incurs substantially lower operational coverage (63.15% vs. 69.42%) and a lower Macro F_1 (93.65% vs. 95.20%). XGBoost is selected as our primary operational architecture because it occupies the superior Pareto-optimal frontier across the multi-objective clinical triage spectrum—maximizing diagnostic discrimination while retaining over 6.2% more usable monitoring windows.

B. Tabular ML (XGBoost) vs. Deep Learning (MLP)

Statistical evaluation using Wilcoxon signed-rank tests and cluster bootstrap differences across the 15 LOSO folds shows:

- **Full Coverage Comparison:** In unconstrained full coverage, Macro F_1 between MLP (88.95%) and XGBoost (88.13%) shows no statistically significant difference ($W = 14.0, p = 0.975, r_{rb} = -0.010, p_{\text{holm}} = 1.0$). However, XGBoost shows lower baseline motion susceptibility ($\text{FAR}_{\text{motion}} = 12.20\%$ vs. 24.00% for MLP).
- **Selective Performance at $\tau = 0.80$:** Both models exceed 95% Selective Macro F_1 (95.38% for MLP,

TABLE II

PERFORMANCE BENCHMARK ACROSS EIGHT MODEL ARCHITECTURES (SEVEN TABULAR ML AND ONE DEEP MLP) UNDER 15-FOLD LOSO CROSS-VALIDATION AT FULL COVERAGE (100%) AND SELECTIVE MODE ($\tau = 0.80$).

Benchmark Model	**Mandatory Prediction (Full Coverage 100%)**				**Post-Calibration**		**Selective Classification ($\tau = 0.80$)**					**FAR Red.**
	Acc (%)	**Macro F_1**	**ECE**	FAR_{mot}	**ECE**	FAR_{mot}	**Coverage**	**Acc (%)**	**Macro F_1**	**ECE**	FAR_{mot}	**(%)**
Logistic Regression	88.93	88.00%	0.0460	23.05%	0.0469	24.41%	72.48%	93.54	92.23%	0.0485	10.17%	55.9%
Decision Tree (CART)	82.43	81.09%	0.1468	22.71%	0.0973	11.86%	50.70%	91.73	85.94%	0.0465	3.05%	86.6%
k-Nearest Neighbors	86.95	85.48%	0.0835	20.34%	0.0548	20.34%	64.94%	95.03	93.69%	0.0500	9.15%	55.0%
AdaBoost Classifier	90.52	89.66%	0.2727	13.90%	0.0722	12.88%	77.13%	94.24	93.29%	0.0548	1.69%	87.8%
Random Forest	89.21	88.01%	0.0638	13.90%	0.0390	16.61%	75.03%	95.32	94.49%	0.0336	3.39%	75.6%
LightGBM	86.53	85.66%	0.0915	13.56%	0.0686	9.49%	63.15%	95.09	93.65%	0.0333	1.36%	**90.0%**
MLP (Deep Learning)	89.63	88.95%	0.0312	24.00%	0.0336	25.63%	75.16%	95.76	95.38%	0.0179	10.17%	57.6%
XGBoost (Proposed)	**88.97**	**88.13%**	**0.0612**	**12.20%**	**0.0744**	**10.85%**	**69.42%**	**96.08**	**95.20%**	**0.0290**	**4.07%**	**66.6%**

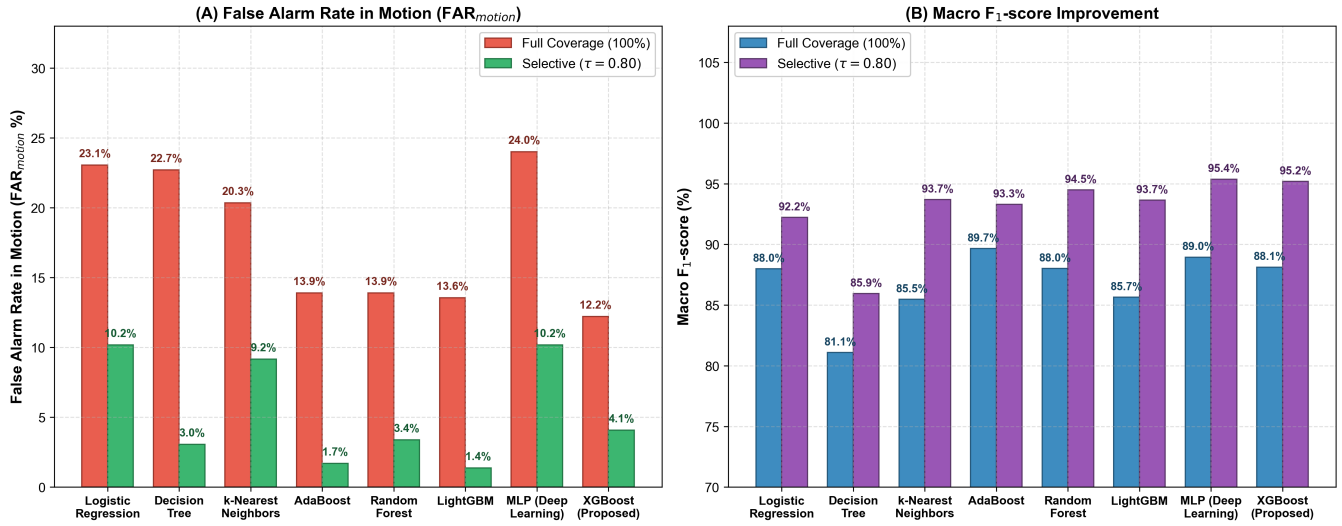


Fig. 1. Cross-model performance comparison across eight algorithms (seven tabular ML + one deep MLP): (A) Motion-induced false alarm rate ($\text{FAR}_{\text{motion}}$) suppressed by 55.0% to 90.0% across all models; (B) Gains in Selective Macro F_1-score at threshold $\tau = 0.80$.

95.20% for XGBoost). The 95% paired bootstrap difference $(MLP - XGB)$ contains zero: $\Delta F_1 = +0.0117$ (95% CI: $[-0.0256, +0.0513]$) and $\Delta\text{FAR}_{\text{sel}} = +2.03\%$ (95% CI: $[-5.44\%, +8.54\%]$), indicating that selective abstention functions consistently across tabular and neural formulations.

- **Subject-Level Stability:** On Subject S14 (evaluated in Fold 12), the MLP captures subtle non-linear cues ($F_{1,\text{stress}} = 97.2\%$), whereas tree algorithms underperform due to suppressed EDA amplitude. Conversely, the MLP exhibits complete abstention on Subject S17 (Fold 15), yielding 0% coverage. In contrast, XGBoost maintains stable coverage across all 15 subjects ($69.42\% \pm 22.2\%$).
- **Wearable Edge Systems Rationale:** From an embedded systems perspective, tree ensembles like XGBoost offer substantial practical advantages over deep end-to-end architectures. While raw-waveform temporal neural networks require intensive floating-point computation, large SRAM allocations, and significant battery drain, our domain-engineered feature extraction pipeline requires only 25 statistical features computed over sliding windows. This feature set can be executed on ultra-low-power microcontrollers (e.g., ARM Cortex-M4/M33 operating under sub-milliwatt power budgets), enabling multi-day continuous ambulatory monitoring on standard wristwear battery capacities.

C. Selective Trade-Offs and Detailed 15-Fold Evaluation (Answering RQ2 & RQ3)

Table III outlines the selective trade-off profile of calibrated XGBoost across thresholds $\tau \in [0.60, 0.90]$. The reduction in motion false alarms is plotted in Fig. 2, probability calibration via Reliability Diagrams is shown in Fig. 3, and the Risk-Coverage curve is presented in Fig. 4.

TABLE III

SELECTIVE CLASSIFICATION TRADE-OFF PROFILE FOR XGBOOST ACROSS CONFIDENCE THRESHOLDS τ UNDER 15-FOLD LOSO VALIDATION.

Configuration	**Coverage**	**Macro F_1**	**Stress F_1**	**ECE**	$\text{FAR}_{\text{motion}}$
XGBoost (Raw Baseline 100%)	100.0%	88.13%	84.99%	0.0612	12.20%
XGBoost + Isotonic (100%)	100.0%	89.88%	86.68%	0.0744	10.85%
Selective ($\tau = 0.60$)	89.76%	92.86%	90.45%	0.0534	7.46%
Selective ($\tau = 0.70$)	81.79%	93.68%	91.28%	0.0406	4.41%
Optimal Selective ($\tau = 0.80$)	**69.42%**	**95.20%**	**93.15%**	**0.0290**	**4.07%**
Conservative Selective ($\tau = 0.90$)	50.38%	96.93%	95.54%	0.0161	0.68%

At the selected operating point $\tau = 0.80$, operational coverage is 69.42% (95% CI: [58.34%, 79.84%]), Selective

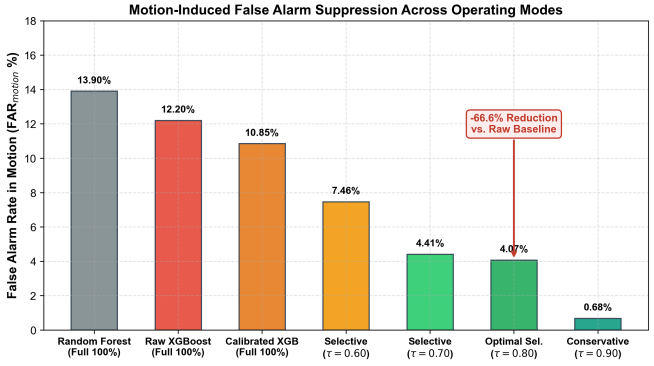


Fig. 2. Suppression of motion-induced false alarm rate ($\text{FAR}_{\text{motion}}$) from mandatory coverage to selective abstention.

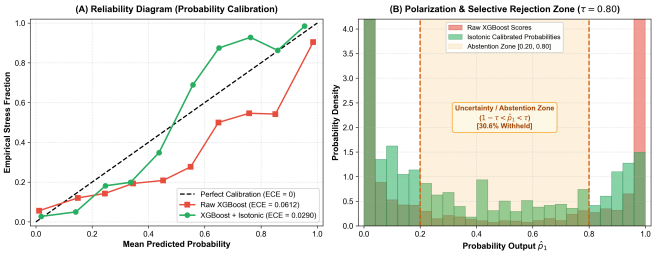


Fig. 3. Reliability diagram: Isotonic calibration aligns XGBoost empirical probability outputs with the ideal diagonal reference.

Macro F_1 reaches **95.20%** (Stress F_1 = 93.15%, 95% CI: [84.80%, 98.54%]), Selective Accuracy is 96.08%, and $\text{FAR}_{\text{motion}}$ decreases to **4.07%** (95% CI: [0.0%, 11.07%]), reflecting a 62.5% reduction compared to the full-coverage calibrated model (10.85%) and a 66.6% reduction relative to the raw baseline (12.20%).

Under 100% coverage, Isotonic regression slightly increases raw ECE from 0.0612 to 0.0744. This occurs because non-parametric piecewise step calibration can produce minor local bin misalignments across subjects during cross-subject LOSO transfer when all windows are retained. However, the primary function of Isotonic calibration in this framework is to serve as a **Confidence Separator**: pushing confident predictions toward 0 or 1 while clustering ambiguous predictions within $[0.3, 0.7]$. When paired with selective rejection ($\tau = 0.80$), this uncertain middle zone is eliminated, lowering ECE to **0.0290**.

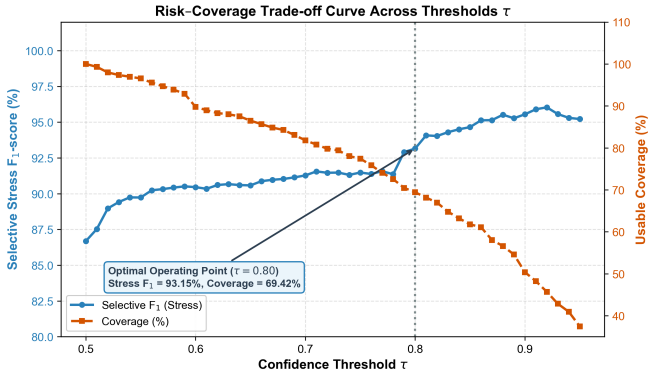


Fig. 4. Risk–Coverage trade-off: As threshold τ increases, selective accuracy and F_1 rise while misclassification error decreases.

Table IV and Fig. 5 report subject-by-subject metrics across all 15 WESAD participants ($S2 \rightarrow S17$, excluding test subject $S1$ and faulty sensor subject $S12$).

Analysis of individual variation in Table IV and Fig. 5 shows:

1) On **8 of the 15 folds** (S4, S6, S10, S11, S13, S15, S16, S17), selective classification achieves $F_{1,\text{sel}} = 100.0\%$, with two additional folds showing high performance (S7: 98.6%, S8: 97.8%).
2) Coverage varies according to sample ambiguity: from 14.8% on Subject S10 to 100.0% on Subject S4. Across all participants, coverage averages $69.6\% \pm 22.2\%$ with average selective accuracy of $96.1\% \pm 7.1\%$. The pronounced drop in coverage for Subject S10 (14.8%) reflects extreme model prudence under high inter-individual physiological ambiguity, choosing to abstain on 85.2% of borderline epochs to safeguard perfect diagnostic precision ($F_{1,\text{sel}} = 100.0\%$, $\text{FAR}_{\text{sel}} = 0.0\%$). In practical continuous ambulatory monitoring, such localized coverage degradation highlights the inherent trade-off of a single population-wide static threshold τ and motivates future investigation into personalized, subject-adaptive confidence thresholds.
3) Selective false alarm rates FAR_{sel} fall to 0.0% across 9 of the 13 subjects with high-motion non-stress windows.

D. Ablation Study: Acceleration Context (Answering RQ4 - Part 1)

To examine whether tri-axial accelerometry is essential or whether physiological channels alone suffice for uncertainty estimation, we conduct an ablation study across five random seeds (42, 123, 456, 789, 2024) using three configurations:

- **Config A (Physio-only):** 19 features (BVP, EDA, TEMP).
- **Config B (Context-aware):** 25 features (Physiological + ACC).
- **Config C (Kinematic-only):** 6 accelerometer features.

Results in Table V and Fig. 6 show:

1) **Consistency Across Seeds:** Adding ACC features (Config B) consistently improves upon physiological features alone (Config A) across seeds. Average Selective Macro F_1 increases from $92.98\% \pm 0.61\%$ to $94.79\% \pm 0.50\%$ (+1.81%), while selective motion false alarms $\text{FAR}^{\text{sel}}_{\text{motion}}$ drop from $4.81\% \pm 0.56\%$ to $3.46\% \pm 0.65\%$ (a 1.35% reduction). In full coverage, ACC features reduce $\text{FAR}^{\text{full}}_{\text{motion}}$ from 13.69% to 10.44% (−3.25%).
2) **Motion Regime Coverage:** In high-motion windows ($\text{Var}(\text{ACC}) \geq \theta_{\text{motion}}$), coverage for Config B is $61.76\% \pm 0.98\%$, compared to $56.05\% \pm 1.94\%$ for Config A. Kinematic signals provide context that helps the classifier retain valid stress epochs that occur during physical activity.

TABLE IV

SUBJECT-LEVEL PERFORMANCE ACROSS 15 LEAVE-ONE-SUBJECT-OUT (LOSO) TEST FOLDS UNDER CALIBRATED XGBOOST WITH SELECTIVE CLASSIFICATION ($\tau = 0.80$), LABELED BY ORIGINAL WESAD SUBJECT IDENTIFIERS (SUBJECT S14 EVALUATED IN FOLD 12; SUBJECT S12 EXCLUDED DUE TO SENSOR DEFECT IN WESAD).

Fold / Subject	Epochs n	Stress Ratio (%)	Motion Ratio (%)	Mandatory Prediction (Full Coverage)			Selective Classification ($\tau = 0.80$)			
				$F_{1,\text{XGB}}$ (%)	$F_{1,\text{Cal}}$ (%)	FAR_{Cal} (%)	Coverage (%)	$F_{1,\text{sel}}$ (%)	Acc_{sel} (%)	FAR_{sel} (%)
Fold 1 (S2)	338	34.3%	18.9%	76.6%	71.8%	0.0%	73.1%	17.6%	88.7%	0.0%
Fold 2 (S3)	341	35.5%	24.0%	73.1%	71.2%	63.6%	59.8%	81.1%	77.0%	54.5%
Fold 3 (S4)	345	34.8%	15.1%	100.0%	100.0%	0.0%	100.0%	100.0%	100.0%	0.0%
Fold 4 (S5)	353	34.3%	5.4%	89.0%	92.1%	78.9%	71.7%	96.0%	97.6%	31.6%
Fold 5 (S6)	351	35.0%	4.6%	81.7%	94.2%	7.1%	53.8%	100.0%	100.0%	0.0%
Fold 6 (S7)	351	34.5%	17.4%	85.5%	90.6%	–	60.4%	98.6%	98.6%	–
Fold 7 (S8)	352	35.8%	24.7%	91.4%	90.9%	0.0%	84.1%	97.8%	99.0%	0.0%
Fold 8 (S9)	350	34.6%	8.6%	69.2%	64.8%	0.0%	87.4%	54.7%	84.3%	0.0%
Fold 9 (S10)	365	37.5%	16.2%	69.2%	98.6%	0.0%	14.8%	100.0%	100.0%	0.0%
Fold 10 (S11)	357	35.9%	32.2%	98.0%	100.0%	0.0%	94.4%	100.0%	100.0%	0.0%
Fold 11 (S13)	354	35.6%	42.1%	90.0%	89.0%	0.0%	63.0%	100.0%	100.0%	0.0%
Fold 12 (S14)	357	35.9%	20.2%	65.3%	24.7%	–	64.1%	0.0%	96.1%	–
Fold 13 (S15)	358	36.3%	30.4%	89.0%	90.8%	6.8%	44.7%	100.0%	100.0%	0.0%
Fold 14 (S16)	356	35.7%	11.5%	100.0%	100.0%	0.0%	95.5%	100.0%	100.0%	0.0%
Fold 15 (S17)	366	37.4%	28.1%	98.2%	89.3%	30.0%	76.5%	100.0%	100.0%	30.0%
Mean ± SD	**5294**	**35.5 ± 1.0%**	**20.0 ± 10.5%**	**85.1 ± 12.0%**	**84.5 ± 19.9%**	**14.3 ± 26.8%**	**69.6 ± 22.2%**	**83.1 ± 32.6%**	**96.1 ± 7.1%**	**6.6 ± 16.8%**

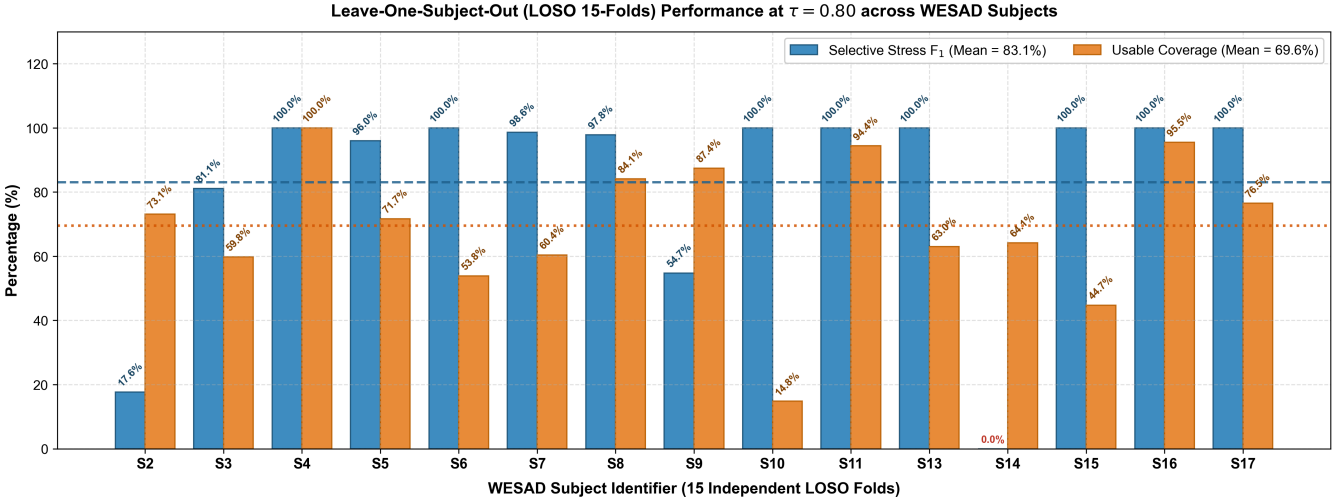


Fig. 5. Subject-level performance distribution across 15 LOSO test folds at $\tau = 0.80$: Selective Stress $F_{1,\text{sel}}$ (blue bars) and Usable Coverage (orange bars) plotted side-by-side for each WESAD participant ($S2 \rightarrow S17$) with dataset averages ($F_{1,\text{sel}} = 83.1\%$, Coverage $= 69.6\%$).

TABLE V

MULTI-SEED ABLATION COMPARISON (5 SEEDS) AT $\tau = 0.80$ ON WESAD LOSO (MEAN ± STANDARD DEVIATION ACROSS RANDOM SEEDS).

Configuration	Coverage (%)	Sel. Macro F_1 (%)	$\text{FAR}^{\text{sel}}_{\text{motion}}$ (%)	$\text{FAR}^{\text{full}}_{\text{motion}}$ (%)
A: Physio-only (No ACC)	68.60 ± 0.71	92.98 ± 0.61	4.81 ± 0.56	13.69 ± 0.74
B: Full Context (+ ACC)	$\mathbf{69.60 \pm 0.33}$	$\mathbf{94.79 \pm 0.50}$	$\mathbf{3.46 \pm 0.65}$	$\mathbf{10.44 \pm 0.44}$
C: Kinematic-only (ACC-only)	45.14 ± 0.93	82.38 ± 0.66	6.10 ± 1.59	26.31 ± 0.70

Note: Binary stress $F_{1,\text{stress}}$ scores at $\tau = 0.80$: Config A: $89.61 \pm 0.87\%$; Config B: $\mathbf{92.58 \pm 0.67}\%$; Config C: $74.23 \pm 1.11\%$.

3) **Statistical Testing Details:** Paired Wilcoxon signed-rank tests across the 15 folds (at seed $S_0 = 42$) produce: $F_{1,\text{sel}}$ ($W = 9.0, p = 0.753, r_{rb} = 0.143$), FAR_{sel} ($W = 1.0, p = 1.000, r_{rb} = -0.333$), and Coverage ($W = 37.0, p = 0.552, r_{rb} = 0.187$), with Holm-Bonferroni corrected $p_{\text{holm}} = 1.000$. Paired cluster bootstrap intervals for difference ($B - A$) span zero: $\Delta F_{1,\text{sel}} = +0.0342$ (95% CI: $[-0.0278, +0.1469]$) and $\Delta\text{FAR}_{\text{sel}} = -1.0169\%$ (95% CI: $[-4.5977, +1.0870]$).

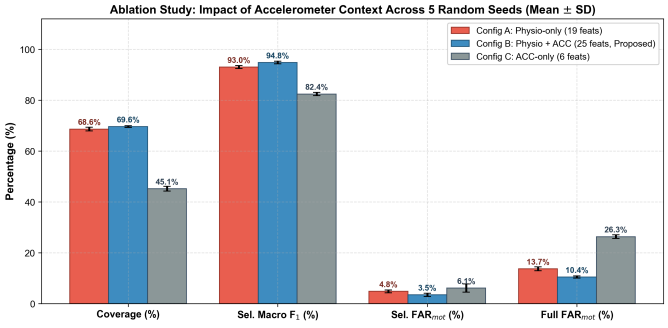


Fig. 6. Multi-metric ablation comparison at $\tau = 0.80$: Accelerometer context increases usable coverage while limiting motion-induced false alarms.

These paired results reflect differences in individual physical activity profiles:

- For active individuals (e.g., Subject S5 where FAR_{sel} drops from 52.6% to 31.6%, or Subject S9 where

$F_{1,\text{sel}}$ rises from 13.8% to 54.7%), accelerometry provides an important baseline to avoid false positives.

- Conversely, for sedentary individuals ($S4, S6, S11$), high signal quality in BVP and EDA renders additional motion features less impactful.

Therefore, kinematic context acts as a helpful safeguard for unconstrained daily life monitoring.

E. TreeSHAP Attribution and Outlier Analysis (Answering RQ4 - Part 2)

1) Global Modality Attribution: TreeSHAP attribution [24] across out-of-fold predictions (shown in Fig. 7) indicates EDA accounts for 31.6% of model attribution, followed by BVP/HRV (26.6%), TEMP (25.9%), and ACC (15.9%). The primary individual features are `eda_tonic_mean` (13.6%), `hrv_mean_hr` (12.3%), `temp_max` (10.6%), and `eda_scr_peaks_count` (10.3%).

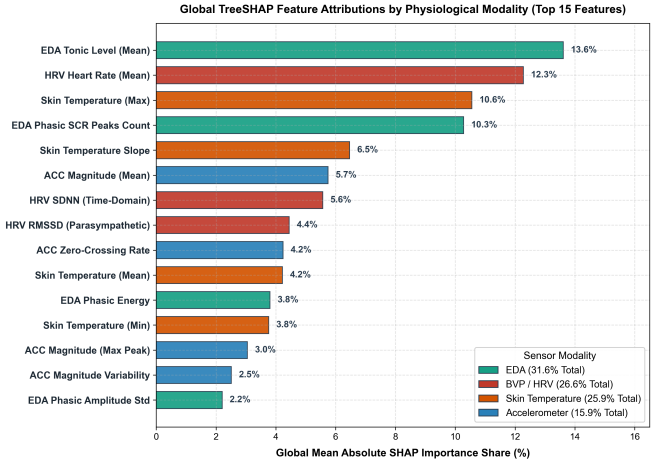


Fig. 7. Global TreeSHAP feature attributions for wrist-based stress classification.

2) Local Abstention Patterns: Examining rejected windows (*Uncertain*) reveals conflicting feature indications: accelerometers register movement (pushing predictions toward non-stress), while heart rate and skin conductance fluctuate ambiguously. This conflict yields intermediate posterior probabilities that trigger abstention.

3) Physiological Dissociation in Outlier Subject S14: Fig. 8 illustrates an outlier case on Subject S14 (evaluated in Fold 12; note that WESAD subjects span $S2 \rightarrow S17$, with $S12$ omitted due to sensor error). Data traces show Subject S14 exhibits traits of an *electrodermal hyporesponder* [10]: Standardized z-scores for `eda_tonic_mean` and `eda_scr_peaks_count` are substantially low (-0.90 and -1.02), resulting in a strong negative SHAP contribution (-3.22) toward Non-Stress that offsets clear tachycardia ($z = +2.76$). This physiological dissociation explains why standard EDA-weighted models struggle ($F_{1,\text{sel}} = 0.0\%$) on individuals with atypical sudomotor patterns, indicating the value of adaptive personalization in future work.

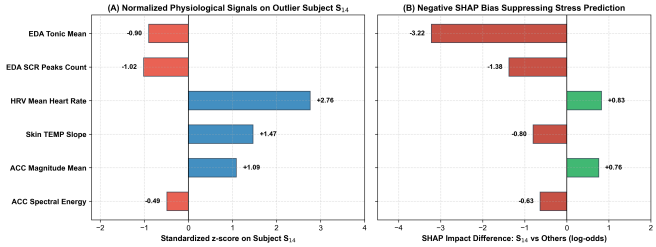


Fig. 8. Physiological dissociation on outlier Subject S14: Low electrodermal reactivity suppresses tachycardia-derived stress classifications.

VI. Conclusion

This study evaluated wrist-based stress detection under mandatory 100% prediction versus selective classification. Across eight model families evaluated under 15-fold LOSO cross-validation, motion-induced false alarms proved widespread under forced prediction, whereas selective classification suppressed these false alerts by 55.0% to 90.0%. The proposed framework yields strong reliability metrics (Selective Macro $F_1 = 95.20\%$, ECE $= 0.0290$, $\text{FAR}_{\text{motion}} = 4.07\%$) while maintaining 69.42% operational coverage. Supported by TreeSHAP interpretability and validation across 15 participants, the approach offers practical design principles for wearable health monitoring.

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